A FLAT GROUP SCHEME OF FINITE TYPE OVER THE AFFINE LINE WHICH IS NOT SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is 'See the discussion above' - the construction is the context span, which is the same material delivered as the context of L48293; content verbatim.

Editorial note. The label lemma-non-separated-group-scheme names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3830-3856: G = affine line with 0 doubled over S = A^1_k, fibres singletons or two-point sets with the U-point as zero, opens U, V mapping isomorphically to S with U cap V -> S \ {0}, multiplication defined piecewise on the four pieces, author's own omitted group-structure verification
- Schemes, example-affine-space-zero-doubled; the diagonal criterion for separatedness

Source reference. Original lemma 047L: sources/stacks-chow/tags.txt, source lines 5512–5512. Prerequisite pointer invoked at 3871; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 01JD: sources/stacks-chow/tags.txt, source lines 2010–2010. Prerequisite pointer invoked at 3877; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Every group scheme over a field is separated, see Groupoids, Lemma 047L. This is not true for group schemes over a base.

Let k be a field. Let $S = \mathrm{Spec}(k[x]) = \mathbf{A}^1_k$. Let G be the affine line with 0 doubled (see Schemes, Example 01JD) seen as a scheme over S. Thus a fibre of $G \to S$ is either a singleton or a set with two elements (one in U and one in V). Thus we can endow these fibres with the structure of a group (by letting the element in U be the zero of the group structure). More precisely, G has two opens U, V which map isomorphically to S such that $U \cap V$ is mapped isomorphically to $S \setminus \{0\}$. Then

$$G \times_S G = U \times_S U \cup V \times_S U \cup U \times_S V \cup V \times_S V$$

where each piece is isomorphic to S. Hence we can define a multiplication $m : G \times_S G \to G$ as the unique S-morphism which maps the first and the last piece

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

into U and the two middle pieces into V. This matches the pointwise description given above. We omit the verification that this defines a group scheme structure.

Lemma 0.1. *There exists a flat group scheme of finite type over the affine line which is not separated.*

Proof. See the discussion above. □